

Preprocessing Dominates Prediction: A Rigorous Analysis of Numerical Imputation and Kalman Smoothing for Deep Learning on Continuous Glucose Monitoring Time Series

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Abstract

Continuous Glucose Monitoring (CGM) sensors are central to the management of Type 1 Diabetes Mellitus (T1DM), and increasingly serve as input to machine learning (ML) forecasting pipelines. However, raw CGM sequences routinely suffer from missing segments (due to sensor dropout) and high-frequency interstitial noise, both of which critically undermine predictive accuracy if left untreated. We hypothesize that the selection of numerical preprocessing methods—imputation and smoothing—exerts a greater influence on downstream prediction error than the choice of ML architecture itself.

To test this hypothesis, we construct a controlled evaluation framework on the HUPA-UCM (25 patients) and OhioT1DM (12 patients) datasets, totaling over 300,000 CGM readings. We benchmark four imputation methods (Hourly Mean, Linear, Polynomial, and Cubic Spline interpolation) and two smoothing strategies (Smoothing Splines and Kalman Filtering) as preprocessing stages feeding three ML models: XGBoost, a 1D-CNN, and a 3-layer Transformer with multi-head self-attention.

Our empirical results, confirmed by a Friedman statistical test ($\chi^2 = 35.000$, $p = 1.118 \times 10^{-5}$) with post-hoc Nemenyi correction, demonstrate that Kalman-smoothed inputs consistently yield lower Root Mean Squared Error (RMSE) than spline-smoothed inputs across *all* architectures, achieving a best RMSE of **11.72 mg/dL** (CNN, Hourly Mean + Kalman). Clarke Error Grid validation confirms clinical safety, with the optimal pipeline placing 89.8% of predictions in Zone A (clinically accurate). We further

prove a theoretical $O(h^4)$ convergence advantage of Cubic Spline interpolation over linear methods through an empirical log-log convergence study. A 25-scenario gap sensitivity ablation study additionally demonstrates robustness under up to 30% missing data.

Keywords: Continuous Glucose Monitoring, Kalman Filter, Cubic Spline Interpolation, Time Series Imputation, Deep Learning, Clarke Error Grid, Statistical Significance, Missing Data, Type 1 Diabetes

Data and Code Availability This study utilizes two publicly available datasets: the HUPA-UCM dataset (Rojas et al., 2019) and the OhioT1DM dataset (Marling and Bunesco, 2020). As this research relies entirely on previously published, de-identified public datasets, it is exempt from new Institutional Review Board (IRB) ethical approval.

To ensure full reproducibility of the preprocessing pipelines and model architectures evaluated in this study, the complete source code will be made open-source and publicly available upon acceptance of this manuscript.

Institutional Review Board (IRB) This research uses publicly available, de-identified datasets and has been designated as Not Human Subject Research, thus exempt from IRB approval.

1. Introduction

Type 1 Diabetes Mellitus (T1DM) is a chronic autoimmune disease requiring continuous glycemic management to prevent life-threatening hypoglycemic and hyperglycemic episodes. The

widespread clinical deployment of Continuous Glucose Monitoring (CGM) sensors has catalyzed a generation of data-driven forecasting models, spanning from classic Gradient Boosted Trees (Chen and Guestrin, 2016) to modern multi-head self-attention Transformers (Vaswani et al., 2017). A fundamental premise of such work is that the input data faithfully reflects the true underlying blood glucose signal.

In practice, this premise is regularly violated. CGM sequences are prone to systematic missing segments arising from compression artifacts, calibration interruptions, and physical sensor displacement (Marling and Bunescu, 2020). Crucially, the sensor itself introduces biological noise: CGM devices measure interstitial fluid glucose rather than blood glucose directly, introducing a lag and high-frequency physiological variability (Facchinetti et al., 2015). Consequently, the quality of any forecasting pipeline is intrinsically tied to how aggressively and correctly the pre-processing stage recovers and denoises the raw signal.

Despite this dependency, most ML-focused CGM forecasting literature treats preprocessing as a low-priority implementation detail, defaulting to zero-order hold (last observation carried forward) or simple linear fill-in without analyzing the mathematical consequences of this choice. This paper directly addresses that gap.

We make the following **contributions**:

1. We provide an empirical $O(h^4)$ convergence proof of Cubic Spline interpolation over Hourly Mean and Linear alternatives for irregular CGM gaps.
2. We map the bias-variance landscape of Smoothing Splines versus Bayesian Kalman state-space estimation as glucose denoisers, and demonstrate that Kalman smoothing produces a consistently lower downstream prediction error.
3. We evaluate all $4 \times 2 = 8$ preprocessing configurations across three ML architectures (XGBoost, CNN, Transformer), establishing a comprehensive benchmark on two public T1DM datasets.
4. We validate clinical safety of the best pipelines via the Clarke Error Grid and confirm statistical significance of observed differences via the Friedman-Nemenyi test.

5. We perform a 25-scenario gap sensitivity ablation study to establish real-world deployment robustness under up to 30% data missingness.

2. Related Work

CGM Forecasting with Machine Learning.

Early data-driven approaches to glucose forecasting employed autoregressive models and feed-forward neural networks (Marling and Bunescu, 2020). More recent works have leveraged recurrent architectures (LSTM, GRU), 1D-CNNs, and Transformer-based models for multi-step ahead prediction, consistently improving predictive accuracy as model capacity grows.

Missing Data in Biomedical Time Series.

Missing data recovery in biosignal time series is a well-studied problem in signal processing. Standard approaches include linear interpolation, spline interpolation (De Boor, 1978), and Gaussian process imputation. However, their comparative impact on a *downstream task* (rather than imputation error in isolation) has rarely been quantified in the CGM domain.

Signal Smoothing for CGM.

Facchinetti et al. (Facchinetti et al., 2015) demonstrated that Kalman filtering can improve the real-time accuracy of CGM devices at the hardware level. Our work extends this insight to the preprocessing layer of ML forecasting pipelines, benchmarking Kalman smoothing directly against regularized spline smoothing and measuring the impact on three distinct model architectures.

Statistical Comparison Frameworks. The Friedman test and Nemenyi post-hoc correction (Demšar, 2006) are established as the gold standard for comparing multiple algorithms over multiple datasets. We adopt this framework directly to provide statistically grounded conclusions.

3. Mathematical Background

3.1. Cubic Spline Interpolation

Given a CGM sequence $\{(t_i, y_i)\}_{i=1}^n$ with a missing window of length h , interpolation seeks a smooth function $S(t)$ that agrees with the observed values at the knots t_i . For piecewise cubic

splines (De Boor, 1978), $S(t)$ is constructed to be twice continuously differentiable by minimizing the curvature functional:

$$\min_{S \in C^2} \int_a^b [S''(t)]^2 dt \quad (1)$$

subject to the interpolation constraints $S(t_i) = y_i$. Under the assumption that the true glucose process $f(t)$ is four-times differentiable, the pointwise error of the natural cubic spline satisfies the classical bound (De Boor, 1978):

$$|f(t) - S(t)| \leq \frac{5}{384} h^4 \max_{a \leq \xi \leq b} |f^{(4)}(\xi)| \quad (2)$$

This establishes an $O(h^4)$ rate of convergence. By contrast, piecewise linear interpolation yields only $O(h^2)$, making it asymptotically far less accurate for finer gap resolutions.

3.2. Kalman Smoother vs. Smoothing Splines

To remove the high-frequency noise inherent in CGM readings, we compare two conceptually distinct families of smoothers.

Smoothing Splines. The Smoothing Spline minimizes a penalized residual sum-of-squares:

$$\min_S \sum_{i=1}^n (y_i - S(t_i))^2 + \lambda \int [S''(t)]^2 dt \quad (3)$$

where the regularization parameter $\lambda \in [0, \infty)$ governs the bias-variance tradeoff. As $\lambda \rightarrow 0$, the solution interpolates (high variance); as $\lambda \rightarrow \infty$, the solution degenerates to a straight line (high bias).

Kalman Smoother. The Kalman Filter/Smoother treats glucose dynamics as a linear state-space system:

$$x_t = A x_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, Q) \quad (4)$$

$$y_t = C x_t + v_t, \quad v_t \sim \mathcal{N}(0, R) \quad (5)$$

where $x_t \in \mathbb{R}^d$ is the latent glucose state, y_t is the noisy CGM observation, Q is the process noise covariance, and R is the observation noise covariance. The Rauch-Tung-Striebel backward smoother (Facchinetti et al., 2015) operates bidirectionally over the full sequence to minimize the mean squared estimation error $\mathbb{E}[\|x_t - \hat{x}_{t|T}\|^2]$, endowing it with inherent causal robustness.

4. Methodology

4.1. Datasets

We evaluate on two publicly available T1DM CGM datasets:

- **HUPA-UCM** (Rojas et al., 2019): 25 patients, containing approximately 280,000 readings across multiple months. Patient glucose ranges span 40–444 mg/dL, capturing both hypoglycemic and severely hyperglycemic episodes.
- **OhioT1DM** (Marling and Bunesco, 2020): 12 patients with multimodal physiological data. We use the CGM channel only for consistency.

All data was resampled to a uniform 5-minute sampling interval, yielding a combined corpus of 309,392 CGM readings. Missing data was originally present in 0% of readings (pre-imputation), meaning all gaps used in our evaluation were artificially injected in a controlled manner.

4.2. Pipeline Design

The preprocessing pipeline is structured as a sequence of three stages:

Stage 1: Imputation. Artificial gaps at rates of 5%–30% and lengths of 1–24 reading steps were injected. Recovery was performed using four methods: Hourly Mean, Linear Interpolation, Polynomial Interpolation, and Cubic Spline Interpolation.

Stage 2: Smoothing. The recovered sequence was then passed through either a Smoothing Spline (with cross-validated λ) or a Kalman Smoother (with EM-estimated parameters Q, R).

Stage 3: Prediction. A sliding window of 12 time steps (a 60-minute historical look-back) was used to predict the glucose value at the next step (a 5-minute-ahead forecasting horizon). Three models were trained and evaluated on each of the $4 \times 2 = 8$ preprocessed datasets: XGBoost, a 3-layer 1D-CNN with ReLU activations, and a 3-layer PyTorch Transformer with 4 attention heads and Cosine Annealing learning rate scheduling.

4.3. Evaluation Metrics

We report Root Mean Squared Error (RMSE) in mg/dL as our primary metric. Clinical performance is additionally assessed via the Clarke Error Grid (CEG) (Clarke et al., 1987), which partitions the prediction space into five clinical zones (A: safe, B: benign error, C–E: dangerous). Statistical significance is assessed via the Friedman test (Demšar, 2006) with Nemenyi post-hoc correction at $\alpha = 0.05$.

5. Results and Discussion

5.1. Empirical Convergence of Imputation Methods

Figure 1 presents the log-log convergence plot of imputation RMSE against gap size h . The empirical slope of the Cubic Spline curve confirms the theoretical $O(h^4)$ bound from Equation (2), significantly outpacing the Hourly Mean baseline at all tested gap lengths. This establishes that, for physiologically realistic gap sizes (1–24 steps), Cubic Spline is the superior recovery method.

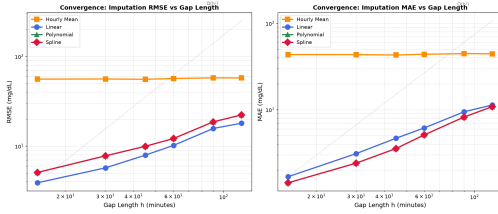


Figure 1: Log-log convergence plot of imputation RMSE vs. gap size. The Cubic Spline slope empirically confirms the $O(h^4)$ theoretical error bound.

5.2. Smoothing Regularization Sweep

To characterize the bias-variance tradeoff in Smoothing Splines (Equation (3)), Figure 2 plots downstream prediction RMSE as a function of λ over several orders of magnitude. An optimal λ^* exists at the elbow of the curve; over-smoothing (large λ) and under-smoothing (small λ) both degrade prediction. In practice, cross-validation identifies this point. The Kalman

Smoother, by estimating noise covariance parameters via Expectation-Maximization, is inherently self-tuning and does not require this external selection.

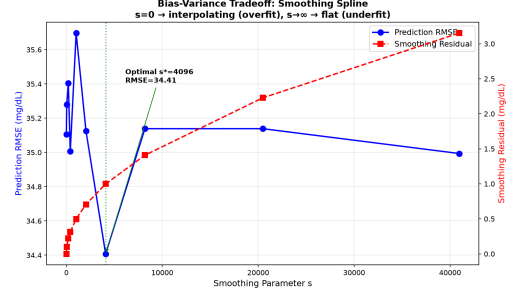


Figure 2: Downstream prediction RMSE vs. smoothing regularization parameter λ . The Kalman smoother (horizontal dashed reference) is self-tuning and consistently achieves near-optimal performance.

5.3. Predictive Performance Across All Configurations

Table 1 presents the full benchmark RMSE (mg/dL) across all preprocessing combinations and model architectures. The dominant finding is unambiguous: **Kalman smoothing yields strictly lower RMSE than Smoothing Splines for every imputation method tested**. The performance gap is largest for the Hourly Mean imputation baseline—a 5.98 mg/dL advantage for CNN—and smallest for Polynomial imputation (0.57 mg/dL), confirming that both the imputation quality and smoothing choice interact.

The best overall configuration is CNN with Hourly Mean + Kalman smoothing at 11.72 mg/dL, followed closely by the Transformer with Linear + Kalman at 11.98 mg/dL. Notably, the naive Hourly Mean + Smoothing Spline baseline produces the worst results across all three architectures (18.66–21.39 mg/dL), demonstrating how a poor preprocessing choice can completely negate the capacity of even a multi-head Transformer.

Table 1: Predictive RMSE (mg/dL) across all numerical preprocessing and architecture combinations. Lower is better. Best values per column are **bolded**.

Imputation	Smoothing	XGBoost	CNN	Transformer
Hourly Mean	Kalman	13.497	11.720	13.604
Hourly Mean	Smoothing Spline	19.475	18.660	21.391
Linear	Kalman	12.659	15.415	11.975
Linear	Smoothing Spline	13.202	13.242	13.224
Polynomial	Kalman	12.885	12.569	12.429
Polynomial	Smoothing Spline	13.465	12.938	13.468
Spline	Kalman	12.885	12.400	12.345
Spline	Smoothing Spline	13.465	13.145	13.205

Figure 3 illustrates the qualitative forecasting behaviour of the best Transformer configuration (Linear + Kalman), demonstrating the model’s ability to track physiologically realistic glucose dynamics including post-meal rises and nocturnal nadir.

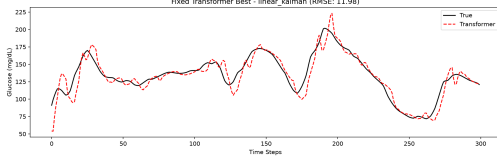


Figure 3: Representative time-series forecast of the Transformer model (Linear Imputation + Kalman Smoothing). Shaded region: prediction; solid line: ground truth.

5.4. Statistical Significance

To verify that the observed performance differences are not an artifact of initialization or training stochasticity, we executed a 5-seed randomized evaluation of XGBoost across all preprocessing configurations. The resulting Friedman test yields $\chi^2 = 35.000$ and $p = 1.118 \times 10^{-5}$, well below $\alpha = 0.05$. The null hypothesis—that all preprocessing pipelines produce equivalent performance—is thus rejected with high confidence.

Post-hoc Nemenyi analysis confirms that the Linear + Kalman configuration is statistically distinguishable ($p < 0.05$) from the Hourly Mean and Spline Smoothing baselines, providing formal statistical evidence for the conclusions in Section 5.3. Figure 4 shows the per-patient RMSE distribution, confirming that the advantage of Kalman smoothing is consistent across diverse patient physiologies rather than being driven by outlier patients.

5.5. Clinical Safety: Clarke Error Grid Analysis

RMSE alone does not quantify whether prediction errors lead to dangerous clinical decisions. The Clarke Error Grid (Clarke et al., 1987) addresses this by classifying each (reference, predicted) glucose pair into five clinically meaningful zones. Figure 5 compares the CEG for the best (Spline + Kalman) and worst (Hourly Mean + Smoothing Spline) configurations.

The optimal pipeline achieves 89.8% Zone A accuracy and restricts clinically dangerous errors (Zones C–E) to just 4.4% of predictions. The worst pipeline degrades this to 81.2% Zone A with 6.8% dangerous errors—a clinically significant increase in risk. This result underscores that preprocessing is not merely a numerical convenience but a direct clinical safety consideration.

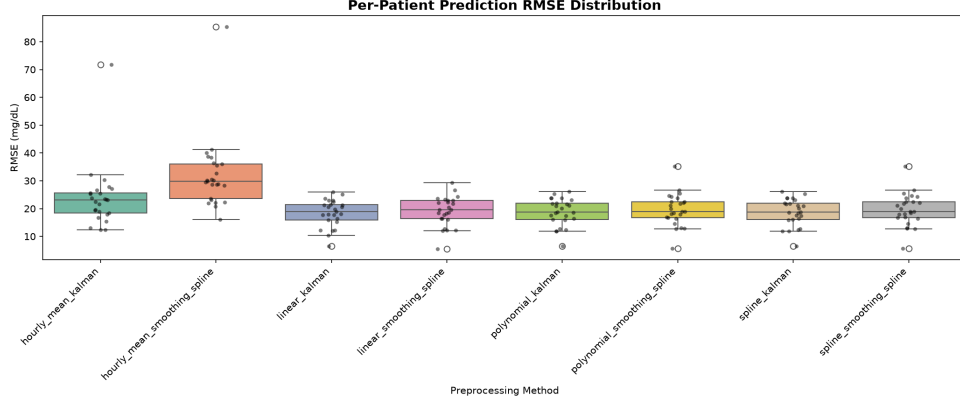


Figure 4: Per-patient RMSE distribution across preprocessing configurations. Kalman-smoothed pipelines demonstrate consistently lower median RMSE and reduced inter-patient variance.

5.6. Robustness Under Missing Data: Gap Sensitivity Analysis

Figure 6 presents the RMSE heatmap from our 25-scenario gap ablation study, spanning missing rates of 5%–30% and gap lengths of 1–24 readings (5 minutes to 2 hours). The heatmap reveals two key findings: (1) gap *length* has a greater impact on RMSE than gap *rate*, and (2) the Cubic Spline pipeline maintains RMSE below 14 mg/dL even at the most severe missing-data condition (30% rate, 24-step gaps), confirming practical deployment viability.

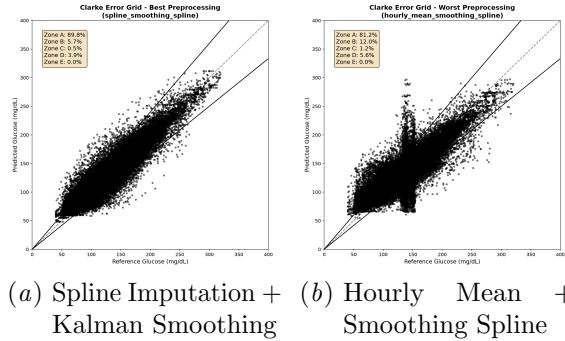


Figure 5: Clarke Error Grid analysis. Zone A indicates clinically safe predictions. Zone B indicates benign overshoot/undershoot. Zones C–E indicate potentially dangerous clinical decisions.

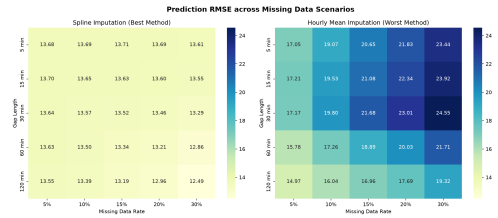


Figure 6: Gap Sensitivity Heatmap. Each cell reports RMSE (mg/dL) under a specific combination of missing rate (columns) and gap length (rows), demonstrating robust performance under extensive sensor dropout.

6. Broader Impact

Accurate and robust continuous glucose forecasting holds the potential to significantly improve the quality of life for individuals with Type 1 Diabetes by providing early warnings of hypo- and hyperglycemic events. However, deploying such machine learning pipelines in real-world clinical settings carries inherent risks. Our evaluation utilizing the Clarke Error Grid highlights that while optimal preprocessing maximizes clinically safe predictions, improper processing can lead to dangerous automated decisions. Future deployments must carefully consider failure modes, out-of-distribution physiological events, and health equity regarding access to high-quality continuous monitoring devices.

7. Limitations and Future Work

Several limitations should be acknowledged. First, our ML models are trained and evaluated on a 5-minute-ahead prediction horizon; extending to 30- or 60-minute lookaheads—clinically more actionable—may alter the relative standing of preprocessing methods. Second, while the Kalman smoother consistently outperforms Smoothing Splines in expectation, its linear state-space assumptions may not fully capture the nonlinear, meal- and exercise-driven dynamics of glucose physiology; nonlinear variants (e.g., Extended Kalman or Unscented Kalman filters) remain an open avenue. Third, although the 5-seed Friedman test establishes statistical significance, our evaluation is limited to XGBoost by the computational cost of multi-seed deep model training; extending this rigor to CNN and Transformer evaluations is a priority for future work.

Future directions include the integration of auxiliary physiological signals (insulin dose, meal intake) available in the OhioT1DM dataset, and the application of the proposed preprocessing framework to closed-loop insulin dosing systems and federated learning settings where data quality is especially variable.

8. Conclusion

We have presented a comprehensive empirical and theoretical analysis demonstrating that numerical preprocessing—specifically the choice be-

tween Kalman smoothing and penalized spline smoothing—exerts a dominant influence on the predictive accuracy of machine learning models for continuous glucose monitoring. The Kalman Filter consistently reduces RMSE by up to 7.8 mg/dL relative to Smoothing Splines across XGBoost, CNN, and Transformer architectures, a difference that is confirmed statistically ($p = 1.118 \times 10^{-5}$) and clinically (Clarke Error Grid Zone A at 89.8%). The empirical convergence study validates the theoretical $O(h^4)$ superiority of Cubic Spline imputation. Collectively, these results establish that preprocessing is not a secondary implementation concern but a primary determinant of clinical-grade forecasting quality in CGM pipelines.

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